

BUILD GUIDE

Bittle X Pi Add-on

A Raspberry Pi Zero 2 W with a battery, a speaker and a microphone, in one rounded box that clicks onto the back of the Petoi Bittle X.

Add-on v0.25
2026-10-26

Platform v3.3
pegs on the stock cover

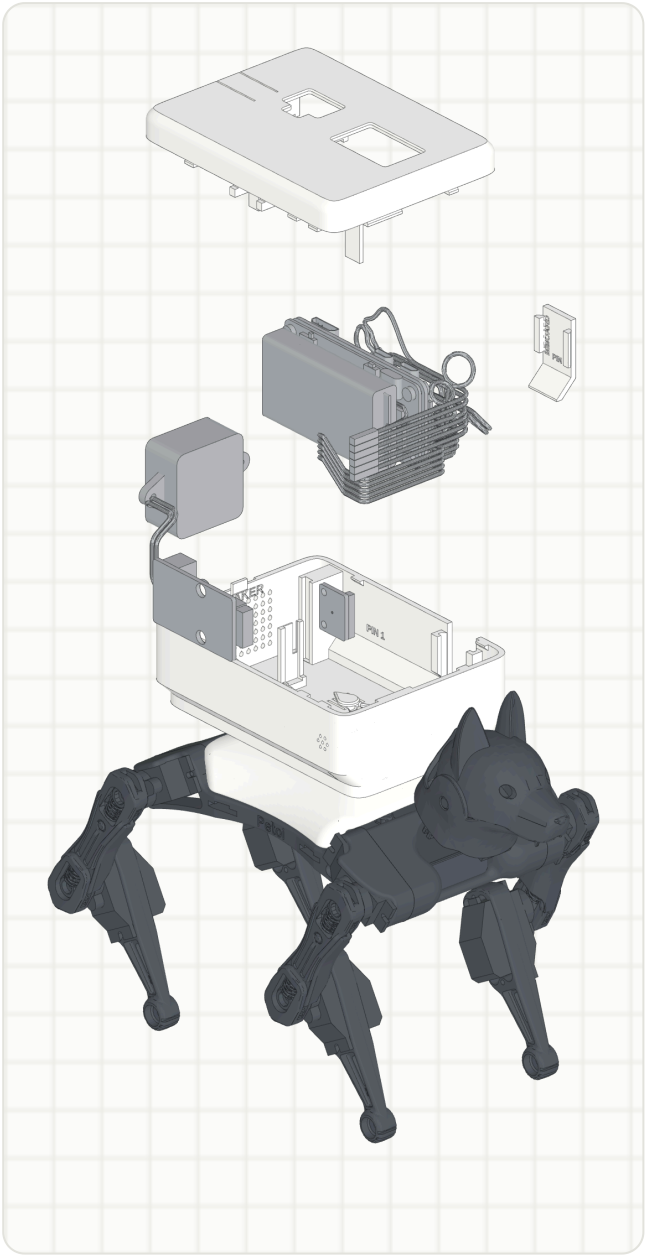
114 g
with the electronics

No supports
for the add-on parts (FDM)

OVERVIEW

What you build

The add-on is a body that clicks onto the platform pegs, a slide-on top panel. Every part goes into the body from the top, in the order of this guide. No tools are necessary, except a soldering iron for the two splices and a dab of glue for the amp.



Directions in this guide

Front is the head end of the robot. Left and right are the robot's left and right. The Pi plugs and the BiBoard header are in the **left well**. The amp and the mic are on the **right wall**. The speaker is at the rear.

Contents

What you build	2
Parts to print	3
Electronics and cables	4
Pin maps and hardware	5
Assembly	6
Wire table	13
Checks, opening and removal	14

The 3D viewer shows every step with animation:
https://beetlebugorg.github.io/bittlex/viewer/

1

Parts to print

Print the add-on parts without supports, in the orientation of the files in stl/print/. The platform cover is in the platform package: print cover_pegs.stl (no foot) with tree supports and a 14-line brim. cover_pegs_foot.stl is the alternative. Its foot teeth leave a spot on the cover where they break off.

PART	MATERIAL	ON THE BED	MASS
Body stl/print/pod_body.stl	PLA, white	floor on the bed	39 g
Top panel stl/print/pod_panel.stl	PLA, second color	upside down: top on the bed	20 g
Door stl/print/pod_door.stl	PLA, white	flat on its outer face	1 g

Settings

0.2 mm layers, 0.25 mm first layer. Walls 2 mm: 4 lines of 0.5 mm. Small holes print about 0.2 mm under size. The fits allow for it.

The panel prints top down: its openings are on the bed. The body prints floor down: its labels and its route grooves face up.

Print plates

stl/plates/ has every part and the platform cover on one bed, in its print orientation. Each 3MF has one object for each part.

PRINTER	FILE	ORIENTATION
FDM: LulzBot TAZ Workhorse, 0.5 mm nozzle	fdm_taz.3mf	As in stl/print/. Use the "Bittle X" Cura profile. The cover stands on its rear end: turn on tree supports and a 14-line brim.
Resin: Peopoly Phenom (first model)	resin_phenom.3mf	The body, the panel and the door are tipped 45 deg. The cover stands on its rear end. Lift the parts 5 mm in the slicer and add supports.

Resin supports

- Body: the floor is toward the plate. Put the supports on the floor underside. Keep them out of the keyhole slots and off the click beams. Sand the marks flat.
- Panel: the open side is toward the plate. The top face has no marks.
- Cover: it stands on its rear end, the points of the peg heads down. Put the supports on the rear edge only.
- Door: the outer face is up, tipped 45 deg. The supports go on the inner face and the ears.
- The resin check found no cups (trapped resin) in these orientations.

Resin fits

- The fits are for FDM. In resin, the holes print closer to size.
- The slide-on fits and the amp slot are a little looser.
- Use a tough resin (ABS-like) for the click beams and the latch tongue.
- Wash the parts in IPA and post-cure them before the fit test.

2

Electronics and cables

Each cable below shows its two ends. Some have one plug at one end and many at the other: match each end before you plug it in.

Amp cable

`1 plug → 5 sockets`

Comes with the NULLLAB NS4168 amp: one 5-pin PH2.0 plug at the amp, five single female Dupont sockets at the other end. The wires are bonded flat. Keep the cable on edge: do not twist it.



BiBoard jumpers

`3 single jumpers` `female → male`

The female end goes onto the Pi pin. The male end goes into the BiBoard socket, which is female.



Mic jumpers

`6 single jumpers` `female → female`

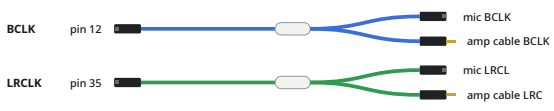
One per mic pin. BCLK and LRCL end in the free female ends of the splices; cut them to length when you make the splices.



Splices

`1 → 2`

Each clock signal feeds two boards. Make a Y from three jumper ends: solder, then cover with white heat shrink.



Speaker lead

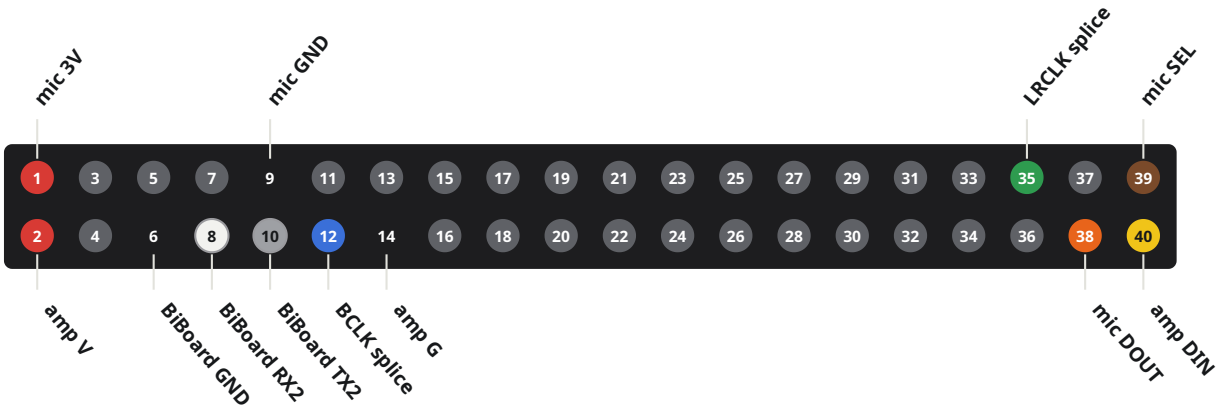
The 2-wire lead of the speaker ends in a 2-pin PH2.0 plug: it goes into the rear end of the amp.

3

Pin maps and hardware

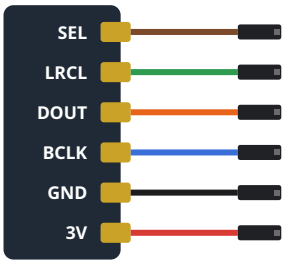
Every pin in use, in the color of its wire.

Pi Zero 2 W, 40-pin header



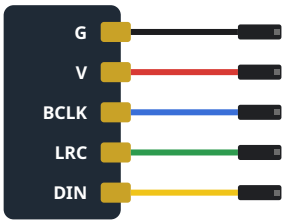
Pin 1 is at the SD card end (rear). Odd pins: inner row (top). Even pins: board edge (bottom).

Mic (as mounted)



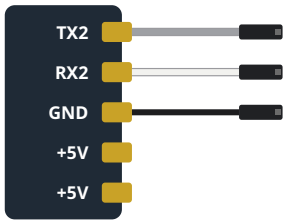
SEL at the top, 3V at the bottom

Amp I2S plug



G at the top (the amp stands upright)

BiBoard socket



male ends in; +5V stays empty

TX and RX cross: BiBoard TX2 goes to Pi RXD (pin 10), BiBoard RX2 to Pi TXD (pin 8). SEL to GND puts the mic on the left channel. The PiSugar S powers the Pi through its contact pins.

Hardware

ITEM	COUNT	USE
Female-to-male Dupont jumper, 10 cm	3 + 2	BiBoard serial link (3) and the amp ends of the splices (2)
Female-female Dupont jumper, 10 cm	4 + 2	mic (4) and the Pi ends of the splices (2)
Female-female Dupont jumper, 15 cm	2	mic GND and 3V
Heat shrink, white	2	splice sleeves
Glue (a dab)		amp

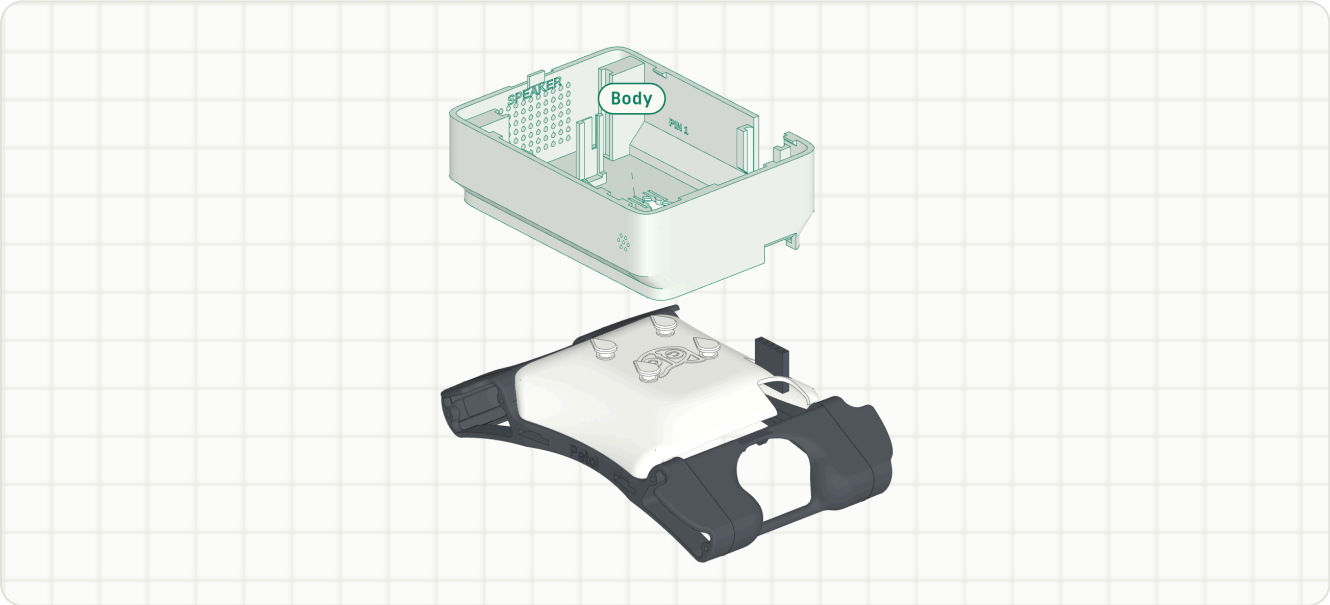
4

Assembly



1 Fit the platform cover

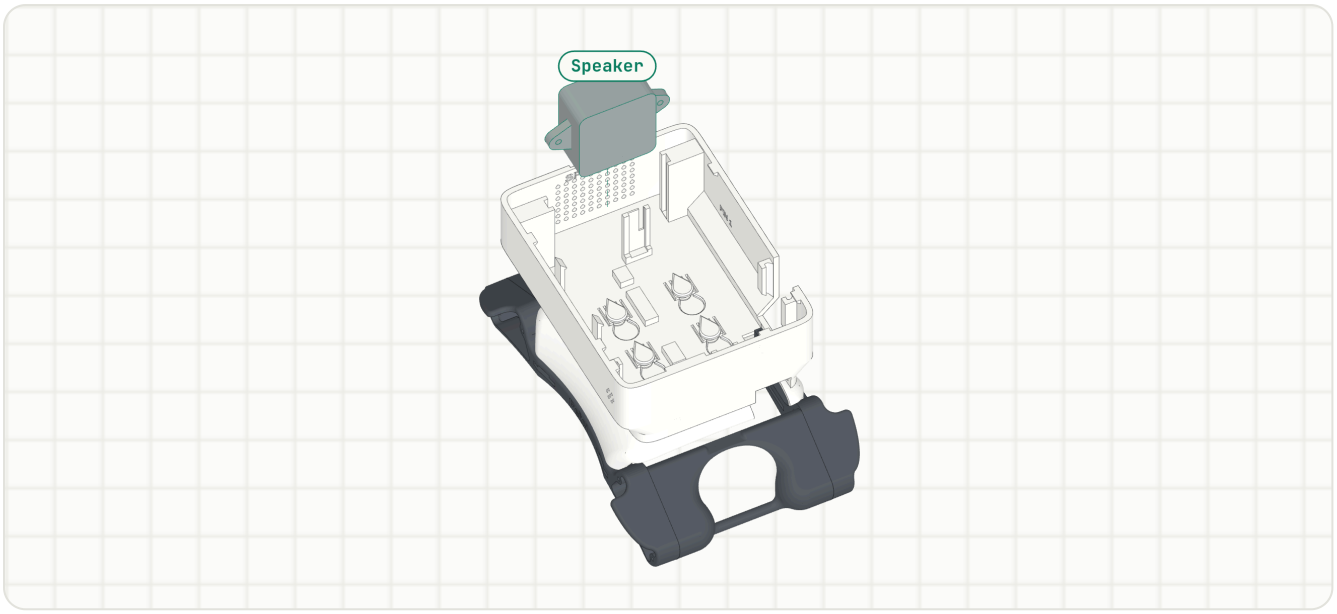
1. Remove the stock back cover of the Bittle X.
2. Put the platform cover in its place. The Pi slot and the BiBoard header are at the front left.
3. Pass the BiBoard header up through the slot.



2 Click the body on

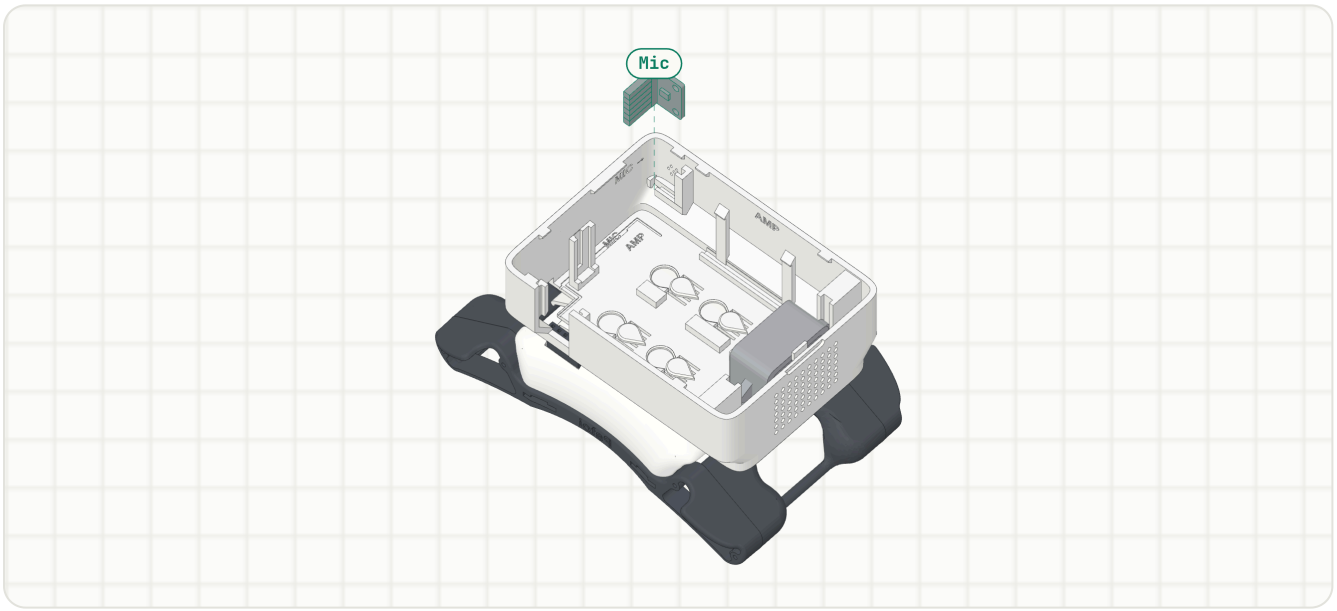
1. Put the empty body on the pegs, 7 mm behind its place, the grille to the rear.
2. Push it 7 mm forward until it clicks (about 2.2 N).
3. The BiBoard header now stands in the notch at the front of the left well.

> Turned the wrong way, the body does not go down onto the pegs.



3 Speaker

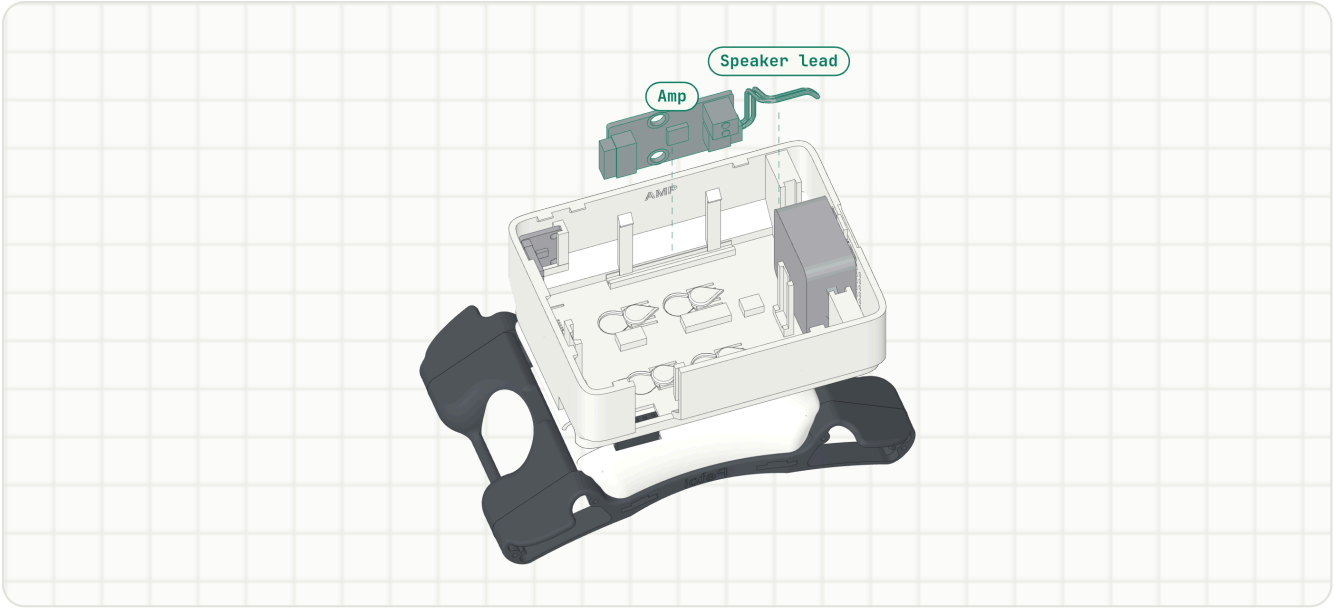
1. Push the two wings of the speaker down into the side grooves at the rear.
2. The speaker body goes to the rear, toward the grille.



4 Mic

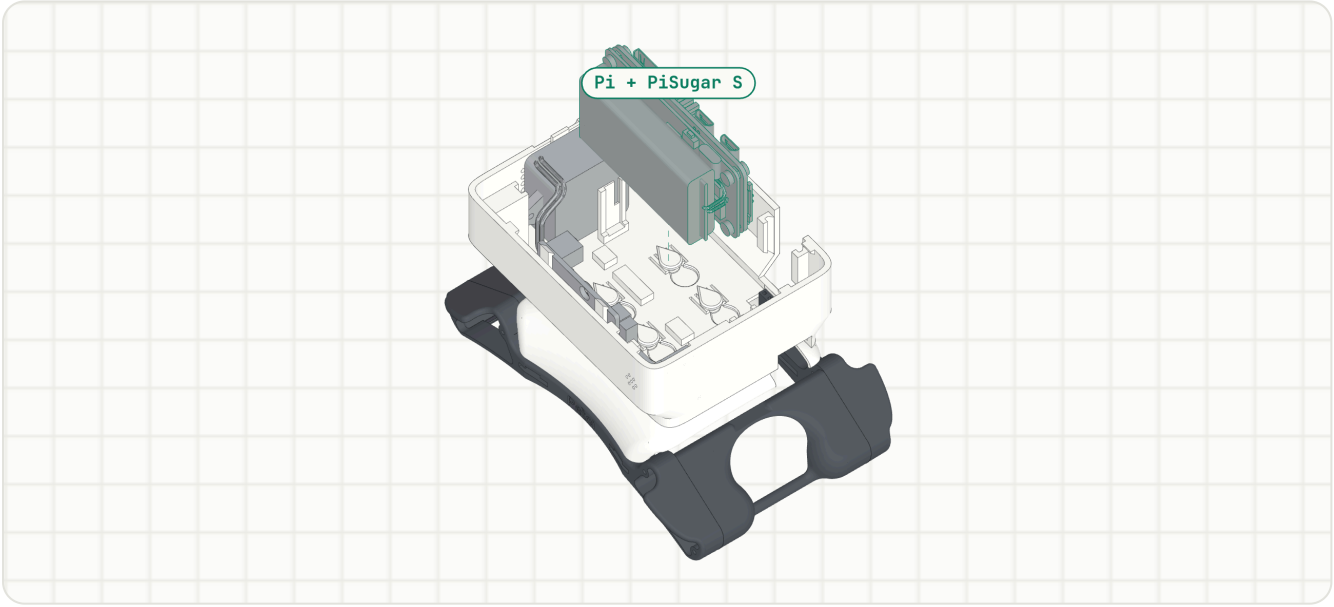
1. Plug the six mic jumpers onto the mic header. Use 15 cm jumpers on GND and 3V.
2. Drop the mic into the slots in the post and the ledge on the right wall, near the front.
3. The port faces the wall. The header is at the front edge, the plugs point into the body.

> The board sits in both slots. The post, the ledge and a stop hold it.



5 Amp and speaker lead

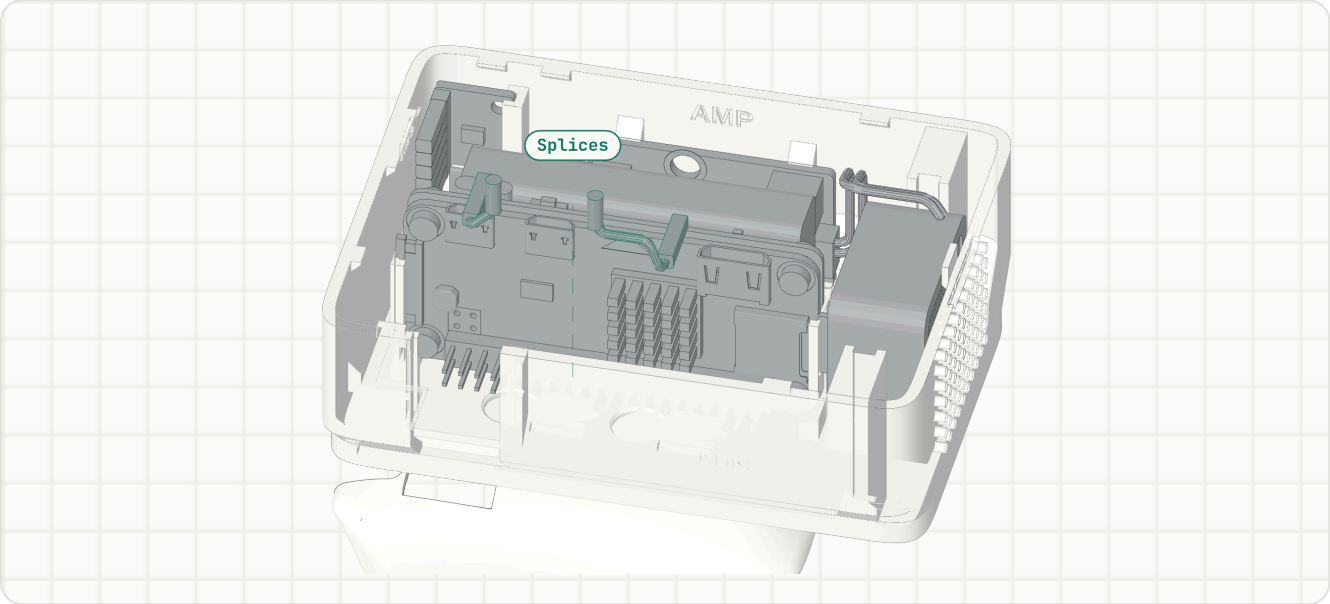
1. Plug the speaker lead into the 2-pin connector at the rear end of the amp.
2. Plug the flat amp cable into the 5-pin connector at the front end. G is at the top.
3. Put the bottom edge of the amp into the slot along the right wall, parts to the inside.
4. Tilt the top edge under the two nubs. Set it with a dab of glue.
5. Two forks under the panel hold the top edge later.
6. Let the amp cable hang out of the top. It goes in after the BiBoard jumpers.



6 Pi + PiSugar S stack

1. Set the PiSugar AUTO switch first: it is under the cell.
2. Push the stack down into the grooves of the two end posts until it stops.
3. The ports are up, the 40-pin header down and toward the left wall, the SD card at the rear.

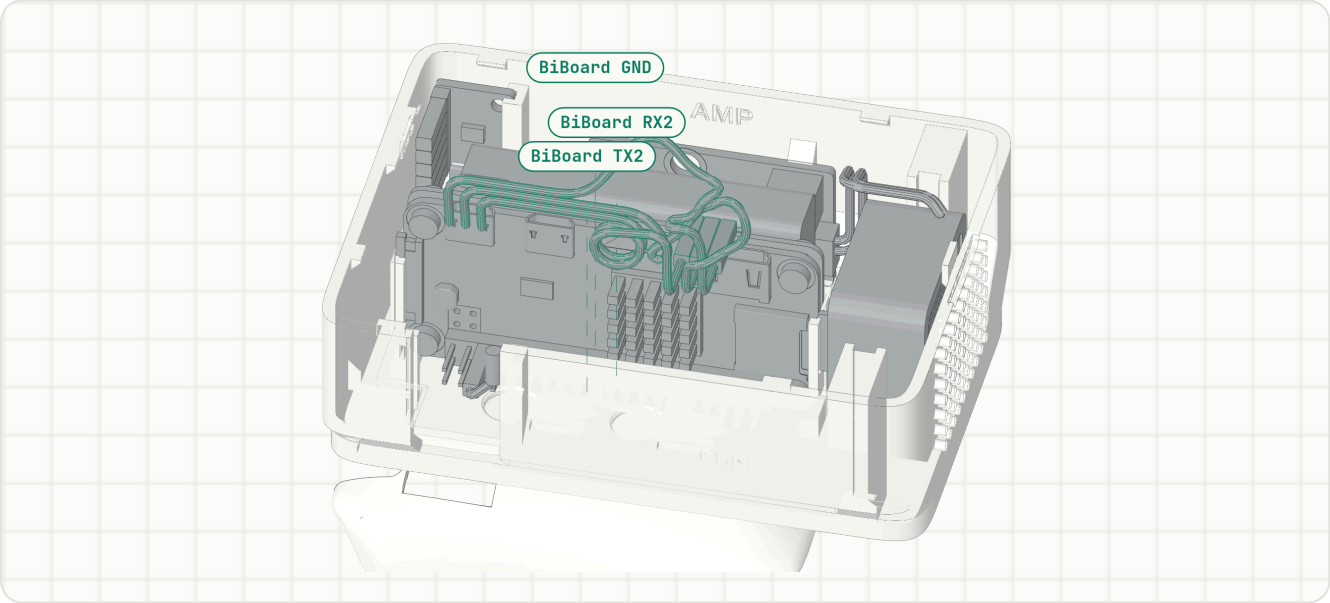
> If the stack is not fully down, the panel does not slide on later.



7 Splices (BCLK and LRCLK)

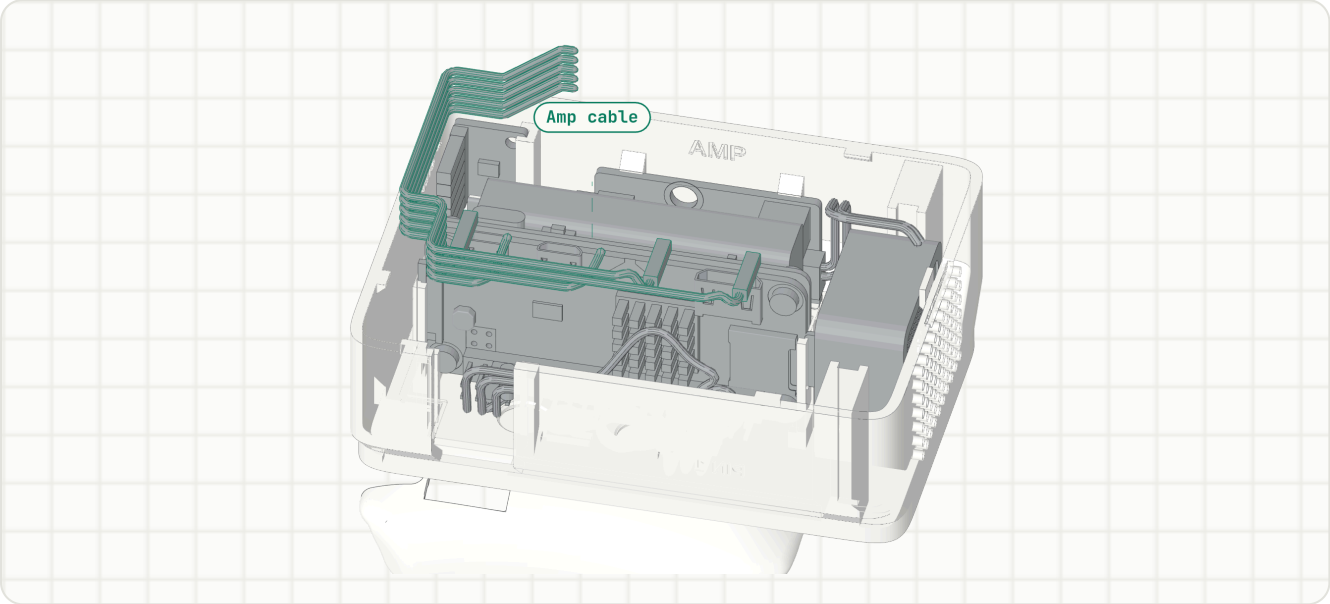
1. Plug the Pi end of each splice: BCLK on pin 12, LRCLK on pin 35.
2. The LRCLK sleeve stands beside the BiBoard plugs, the BCLK sleeve at the middle of the left well.

> Each splice is a Y: one Pi end, one mic end, one amp end. See the cable cards.



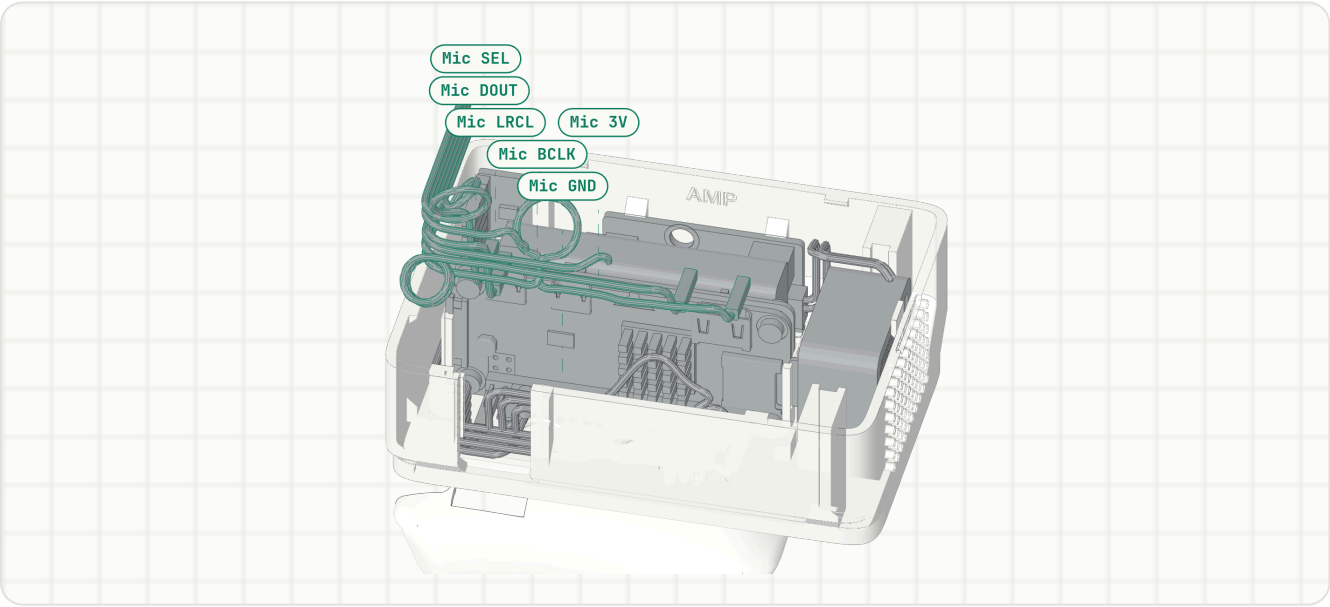
8 BiBoard jumpers: Pi ends

1. Push the female end of each BiBoard jumper onto the Pi: TX2 (gray) on pin 10, RX2 (white) on pin 8, GND (black) on pin 6.
2. Lay the excess of each one in a loop by the left wall.
3. Leave the male ends free. They go onto the BiBoard last.



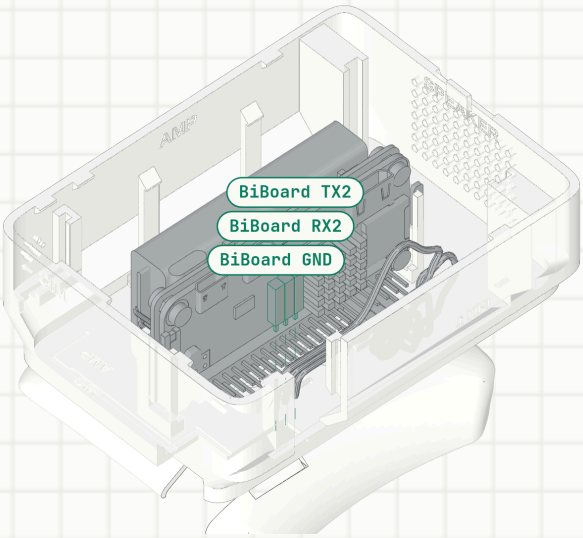
9 Amp cable

1. Lay the flat amp cable on edge, as it leaves the plug, G on top. Do not twist it.
2. It steps down, runs forward to the front wall and across under the mic plugs. Past the front post it turns rearward, runs across behind the front-left notch, and turns rearward along the left well.
3. Peel off each wire at its own height: DIN onto pin 40, LRC and BCLK into the free sockets of their splices, G onto pin 14, V onto pin 2.



10 Mic jumpers

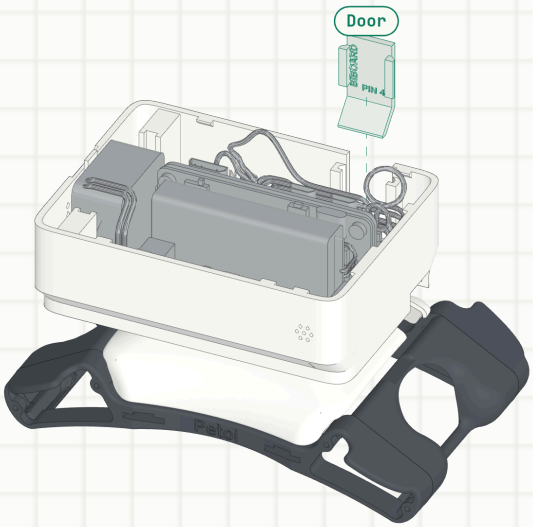
1. Run the mic jumpers straight across behind the front wall, then rearward along the left well.
2. Plug them from the lowest up: 3V on pin 1, GND on pin 9, BCLK and LRCL into their splices, DOUT on pin 38, SEL on pin 39.



11 BiBoard ends

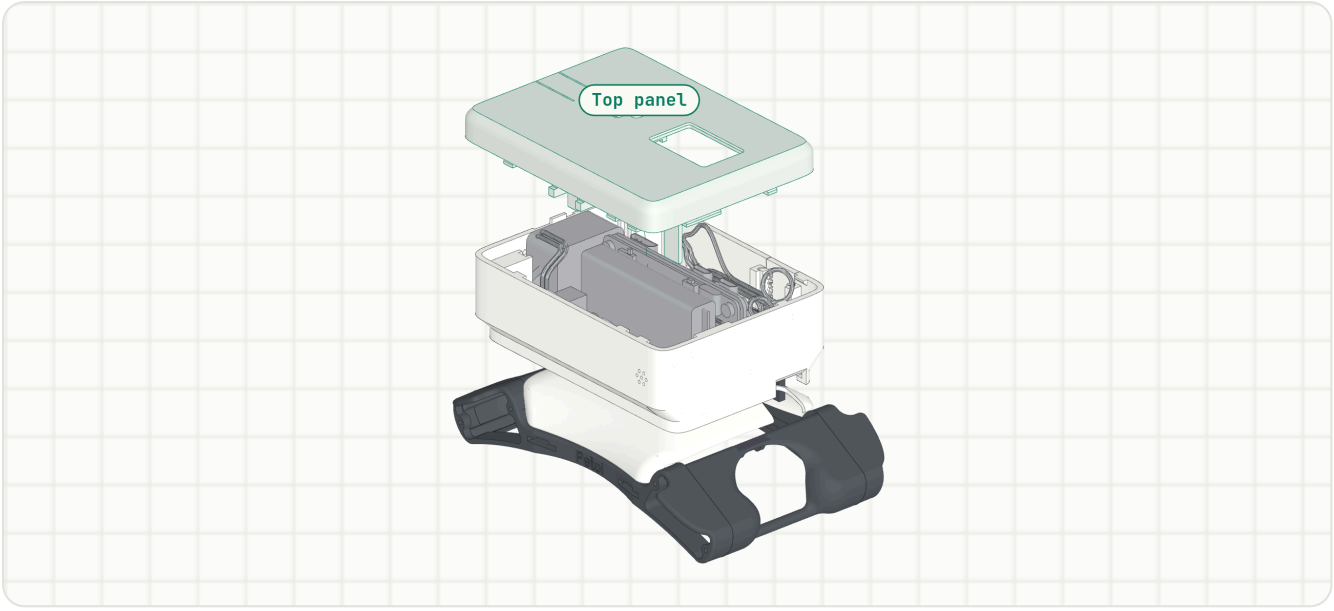
1. Push the male ends of the three BiBoard jumpers into the BiBoard socket: TX2, RX2, GND.
2. Reach the plugs from the side, through the door opening in the left wall.
3. The two +5V sockets stay empty: the PiSugar powers the Pi.

> The other wires are not shown here, for a clear view into the front-left corner.



12 Door

1. Slide the door down into the two grooves of the opening in the left wall. The labels face in.
2. It stands on the wall under it. The panel holds it down.



13 Panel

1. Put the panel down 5 mm behind its place.
2. Slide it forward until the latch at the rear clicks (about 9.2 N).
3. The PiSugar power switch is in the HDMI opening, beside the plug. Move it with a pen tip or a toothpick.
4. Press the PiSugar button through the large opening with a fingertip or a pen tip.

> To open: lift the rear edge at the finger notch and slide the panel rearward.

5

Wire table

Every wire, from end to end. Lengths are between the housings, as modeled; a jumper longer than the route lays its excess in the loop shown.

FROM	TO	GROUP	LENGTH	EXCESS
BiBoard TX2	pin 10	uart	100 mm	2-turn loop
BiBoard RX2	pin 8	uart	98 mm	arch
BiBoard GND	pin 6	uart	98 mm	arch
amp G	pin 14	amp	116 mm	
amp V	pin 2	amp	130 mm	
amp BCLK	BCLK splice	amp	109 mm	
amp LRC	LRCLK splice	amp	95 mm	
amp DIN	pin 40	amp	81 mm	
BCLK splice	pin 12	splice	16 mm	
LRCLK splice	pin 35	splice	7 mm	
mic 3V	pin 1	mic	148 mm	1-turn loop on a 5 mm stem
mic GND	pin 9	mic	149 mm	2-turn loop
mic BCLK	BCLK splice	mic	76 mm	cut to length
mic DOUT	pin 38	mic	97 mm	1-turn loop on a 5 mm stem
mic LRCL	LRCLK splice	mic	62 mm	cut to length
mic SEL	pin 39	mic	99 mm	1-turn loop
amp speaker +	speaker +	speaker	43 mm	
amp speaker -	speaker -	speaker	44 mm	

6

Checks, opening and removal

What the model checks

- No printed part and no board touch, except where a fit is meant to touch.
- Every part goes in from the top, in the order of this guide.
- No two wires pass through each other. No designed bend is tighter than 2 mm.
- The head turns from -85 to 85 deg without contact.
- Platform click: about 2.2 N on, 2.2 N off. Panel latch: about 9.2 N.
- Mass 114 g; center of mass 8.8 mm from the robot center.

Open the panel

Lift the rear edge at the finger notch (about 5 N), slide the panel 5 mm rearward and lift it.

Remove the add-on

Open the panel. Lift the door out of the left wall. Pull the three BiBoard plugs off from the side. Slide the body 7 mm rearward and lift it.

Change the SD card

Open the panel and lift the speaker out of its grooves; the card is at the rear end of the Pi.

Files

stl/print: the parts to print. stl/plates: the print plates. cad/: STEP files. README.md: the full reference. Designs CC BY 4.0, code MIT; see LICENSE.md and ATTRIBUTION.md.